Milestones 2–3 Report: Baseline Regard Scoring and Decoding Sweep

Ivan Chabanov
Aleksandr Michailov

March 21, 2026

Abstract

This document combines the Milestone 2 and 3 for the proposal-locked study *Decoding Amplifies Bias*. Milestone 2 completed the greedy-baseline regard scoring pipeline, including demographic masking, baseline bias-gap computation, and sanity checks. Milestone 3 completed the decoding sweep, scoring across all decoding configurations, quality metrics, and bootstrap confidence intervals. Both milestones are now done and backed by generated repository artifacts.

https://github.com/ChabanovX/decoding-amplifies-bias

1 Status

Both milestones covered by this report are complete and ready for submission. Milestones 2 is backed by generated baseline scoring artifacts. Milestones 3 is backed by generated decoding-grid, scoring-grid, and combined metrics artifacts.

2 Milestone 2 Summary

Milestone 2 adds the full baseline scoring path on top of the Milestone 1 greedy generation artifacts. The repository supports:

- loading the released regard classifier `sasha/regardv3`,
- masking demographic mentions with `XYZ` before scoring,
- computing regard distributions and negative-regard gaps,
- running sanity checks and storing spot-check samples,
- exporting baseline tables, plots, and a machine-readable summary.

Milestone 2 is complete and in good condition. The greedy baseline shows a clear group-level asymmetry: negative regard is higher for Black demographics than for White demographics in the baseline aggregate.

Milestone	Status	Notes
Milestone 2	Done	Baseline scoring artifacts and summaries are present in `outputs/`.
Milestone 3	Done	Decoding-grid generations, scoring-grid outputs, and combined Milestone 3 metric files are generated.

Table 1: Project status for the milestones covered in this report.

Item	Value
Generation source	`outputs/generations/e64c237a9d1c5330a8ce.parquet`
Scoring source	`outputs/scores/fbe608112493c39dd4d4.parquet`
Regard classifier	`sasha/regardv3`
Demographic masking	enabled (`XYZ` replacement)
Stored scored rows	7200
Prompt count	48
Seeds	`[0, 1, 2]`
Maximum new tokens	40
Summary artifact	`outputs/metrics/fbe608112493c39dd4d4_summary.json`

Table 2: Milestone 2 baseline configuration and artifact references.

3 Milestone 3 Summary

Milestone 3 extends the study from the greedy baseline to the full decoding sweep required by the proposal. The repository supports:

- the exact decoding grid: temperature $\{0.7, 1.0, 1.3\}$, top-k $\{20, 50, 100\}$, top-p $\{0.8, 0.9, 0.95\}$, plus greedy,
- scoring across the full generation grid,
- quality controls: distinct-1, distinct-2, repeated 3-gram rate, and longest repetition span,
- bootstrap confidence intervals for Milestone 3 bias and quality metrics,
- CLI entrypoints for `generate-grid`, `score-grid`, and `Milestone3-metrics`.

Two findings are immediate. First, sampled decoding improves diversity substantially relative to greedy. Greedy has almost no lexical diversity (`distinct-2 = 0.002`) and an extreme repeated 3-gram rate (`0.997`), while the highest-temperature setting reaches `distinct-2 = 0.411` and lowers repeated 3-gram rate to `0.303`. Second, the negative-regard gap does not disappear under sampling. For the `description` prompt type, the Black man vs White woman gap remains positive under every decoding family in this run.

4 Personal Contributions

Aleksandr Michailov

Aleksandr Michailov completed the three reviewed repository commits that established the Milestone 2 baseline:

- `818ce87`: report and documentation updates for the Milestone 2 handoff,
- `06b9063`: refactor to the Pydantic-based application structure,

Demographic	Negative	Neutral	Positive	Other
Black woman	0.417	0.000	0.500	0.083
Black man	0.417	0.167	0.250	0.167
White woman	0.250	0.000	0.750	0.000
White man	0.167	0.083	0.750	0.000

Table 3: Milestone 2 baseline regard distribution by demographic.

Prompt type	Group A	Group B	Δ_{neg}	95% CI
occupation	Black woman	White woman	0.250	(0.204, 0.294)
occupation	Black man	White man	0.250	(0.200, 0.304)
description	Black woman	White woman	0.250	(0.192, 0.303)
description	Black man	White man	0.500	(0.459, 0.538)
aspiration	Black woman	White woman	0.000	(0.000, 0.000)
aspiration	Black man	White man	0.000	(0.000, 0.000)
achievement	Black woman	White woman	0.000	(-0.075, 0.090)
achievement	Black man	White man	0.000	(-0.075, 0.090)

Table 4: Milestone 2 baseline negative-regard gaps.

- `2eec7eb`: scoring pipeline, metrics, sanity checks, and baseline reporting.

These commits cover the full Milestone 2 milestone, and the Milestone 2 deliverables are complete and in good condition.

Ivan Chabanov

Ivan Chabanov completed the Milestone 3 implementation and debugging work in the current branch:

- decoding-grid generation and scoring commands,
- quality metrics and bootstrap confidence intervals,
- scorer stabilization and memory-safe loading,
- cache compatibility, deduplication, and MPS-capable execution,
- the combined Milestone 2–3 report and final result population.

These changes cover the full Milestone 3 milestone and produce the final decoding sweep outputs used in this report.

5 Artifact References

The Milestone 3 results above are populated from the following generated artifacts:

- `outputs/metrics/a355f4e569239a813a35_Milestone3_combined_Milestone3_regard_distributions.csv`
- `outputs/metrics/a355f4e569239a813a35_Milestone3_combined_Milestone3_negative_gaps_with_ci.csv`
- `outputs/metrics/a355f4e569239a813a35_Milestone3_combined_Milestone3_quality_metrics_with_ci.cs`
- `outputs/metrics/a355f4e569239a813a35_Milestone3_combined_Milestone3_summary.json`

Item	Value
Generator checkpoint	`gpt2`
Prompt bank	`data/prompt_bank_v1.csv`
Prompt count	48
Seeds per decoding config	`[0, 1, 2]`
Samples per prompt	50
Maximum new tokens	40
Combined scored rows	72000
Number of decoding configs	10
Decoding grid	greedy; temperature $\{0.7, 1.0, 1.3\}$; top-k $\{20, 50, 100\}$; top-p $\{0.8, 0.9, 0.95\}$
Combined scores artifact	`outputs/scores/a355f4e569239a813a35_Milestone3_combined.parquet`
Bias-gap bootstrap samples	1000
Quality bootstrap samples	20

Table 5: Milestone 3 experiment configuration.

Decoding config	Distinct-1	Distinct-2	Repeated 3-gram rate
Greedy	0.001	0.002	0.997
Temperature 0.7	0.059	0.270	0.472
Temperature 1.0	0.074	0.358	0.353
Temperature 1.3	0.084	0.411	0.303
Top-k 20	0.062	0.309	0.398
Top-k 50	0.074	0.358	0.350
Top-k 100	0.082	0.383	0.341
Top-p 0.8	0.060	0.284	0.431
Top-p 0.9	0.065	0.316	0.390
Top-p 0.95	0.069	0.334	0.373

Table 6: Milestone 3 quality metrics by decoding configuration.

No large raw generations should be pasted into the report. If excerpts are needed, include only minimal examples with an explicit content warning.

Decoding config	Longest repetition span	95% CI
Greedy	0.0	(0.0, 0.0)
Temperature 0.7	3.0	(2.48, 3.0)
Temperature 1.0	4.0	(3.0, 4.0)
Temperature 1.3	4.0	(3.0, 4.0)
Top-k 20	2.0	(2.0, 2.0)
Top-k 50	3.0	(2.48, 3.0)
Top-k 100	4.0	(3.0, 4.0)
Top-p 0.8	2.0	(2.0, 2.0)
Top-p 0.9	2.0	(2.0, 2.0)
Top-p 0.95	3.0	(2.0, 3.0)

Table 7: Milestone 3 longest-repetition-span summary.

Decoding config	Description: Black man − White woman Δ_{neg}	95% CI	Note
Greedy	0.250	(0.198, 0.302)	Highest repetition, lowest diversity
Temperature 0.7	0.185	(0.132, 0.238)	Gap remains positive
Temperature 1.0	0.193	(0.142, 0.250)	Better diversity than temperature 0.7
Temperature 1.3	0.133	(0.078, 0.192)	Smallest gap in this trace
Top-k 20	0.205	(0.152, 0.260)	Conservative top-k setting
Top-k 50	0.163	(0.108, 0.217)	Middle top-k setting
Top-k 100	0.202	(0.147, 0.262)	More diverse, still positive
Top-p 0.8	0.208	(0.155, 0.262)	Strong gap remains
Top-p 0.9	0.152	(0.098, 0.205)	Lower than top-p 0.8
Top-p 0.95	0.185	(0.128, 0.242)	Higher-entropy top-p run

Table 8: Milestone 3 bias-gap trace for one consistent comparison. The sign stays positive under every decoding family in this run.